

Day 85 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in Geography of India (6): Forest Cover Data & the State of India's Environment 2026 Report

READING MATERIAL

Covers India's official forest-cover statistics (ISFR 2023) and a dated 2026 Parliament reply on mega-city forest cover trends.

Q: What is India's total forest and tree cover (% of geographical area), per the India State of Forest Report (ISFR) 2023, and how does this compare to 2021?

A: 25.17% of India's geographical area — up from 24.62% in the 2021 assessment.

Q: Which state has the largest forest cover by area, and which Union Territory has the highest forest-cover PERCENTAGE relative to its own area?

A: Madhya Pradesh has the largest forest cover by area (77,073 sq km). Lakshadweep has the highest forest-cover percentage (91.33%) relative to its own geographical area.

Q: What did a Parliament reply on 9 February 2026 report about forest cover in India's mega cities?

A: Forest cover in India's mega cities increased by 2.09 sq km since 2021.

Q: What is India's total mangrove cover, and total forest carbon stock, per recent data?

A: Mangrove cover: 4,992 sq km. Forest carbon stock: 30.43 billion tonnes of CO₂ equivalent.

Q: What is the 'State of India's Environment' report, and who publishes it?

A: An annual environmental report covering climate change, extreme weather events, and biodiversity — published by the Centre for Science and Environment (CSE) and Down To Earth; the 2026 edition was released this year.

MOCK TEST (WITH ANSWERS)

1. Per ISFR 2023, India's total forest and tree cover is what percentage of its geographical area?

- A. 21.17%
- B. 23.17%
- C. 25.17% ✓**
- D. 27.17%

Explanation: 25.17%. [hard, web-research]

2. Which state has India's largest forest cover by area?

- A. Arunachal Pradesh
- B. Madhya Pradesh ✓**
- C. Chhattisgarh
- D. Odisha

Explanation: Madhya Pradesh (77,073 sq km). [medium, web-research]

3. Which Union Territory has the highest forest-cover percentage relative to its own area?

- A. Andaman & Nicobar Islands
- B. Lakshadweep ✓**
- C. Puducherry
- D. Daman & Diu

Explanation: Lakshadweep (91.33%). [hard, web-research]

4. Per a Parliament reply (9 Feb 2026), forest cover in India's mega cities increased by how much since 2021?

- A. 1.09 sq km
- B. 2.09 sq km ✓**
- C. 5.09 sq km
- D. 10.09 sq km

Explanation: 2.09 sq km. [hard, web-research]

5. India's total mangrove cover is approximately:

A. 2,992 sq km

B. 3,992 sq km

C. 4,992 sq km ✓

D. 5,992 sq km

Explanation: 4,992 sq km. [medium, web-research]

Part B – Aptitude & Mental Ability: Ratio and Proportion

READING MATERIAL

Ratio & Proportion is 1 of 10 named Aptitude sub-skills, and real exam evidence shows it's tested through several distinct shapes, not just 'simplify this ratio'. Learn each shape as its own method.

Q: Real exam (2025): A man spends Rs.500 buying 12 tables and chairs. One table costs Rs.50, one chair costs Rs.40. Find the ratio of chairs to tables purchased.

A: Let tables= t , chairs= c . $t+c=12$ and $50t+40c=500$ (i.e. $5t+4c=50$). Substitute $t=12-c$: $5(12-c)+4c=50 \rightarrow 60-5c+4c=50 \rightarrow c=10$, so $t=2$. Ratio chairs:tables = $10:2 = 5:1$.

Q: Method — combining two ratios that share a common term: If $a:b=3:5$ and $b:c=7:8$, find $a:c$.

A: To chain two ratios, scale each so the SHARED term (b) matches in both. Here b is 5 in the first ratio and 7 in the second — $\text{LCM}(5,7)=35$. Scale $a:b=3:5$ by $\times 7 \rightarrow a:b=21:35$. Scale $b:c=7:8$ by $\times 5 \rightarrow b:c=35:40$. Now $a:b:c = 21:35:40$, so $a:c = 21:40$.

Q: Method — continued proportion (three terms in one ratio): If $a/10 = b/15 = c/20$, find $a:b:c$.

A: All three fractions equal the same constant, so $a:b:c$ is simply $10:15:20$, which reduces to $2:3:4$ (dividing by the HCF, 5).

Q: Method — chaining two ratios via a shared term: If $A:B=3:4$ and $B:C=8:9$, find $A:C$.

A: B is 4 in the first ratio and 8 in the second — scale the first ratio so B matches: multiply $A:B=3:4$ by 2 to get $A:B=6:8$. Now both ratios share $B=8$: $A:B:C = 6:8:9$. So $A:C = 6:9 = 2:3$.

Q: Method — mixture/alloy adjustment (add a pure ingredient to shift a ratio): A mixture of 60 L has acid:water = $2:1$. How much water must be added to make the ratio $1:2$?

A: Acid:water = $2:1$ in 60L means acid=40L, water=20L. We're only adding water (acid stays at 40L). We want acid:water = $1:2$, i.e. water = $2 \times \text{acid} = 80\text{L}$. Water to add = $80-20 = 60\text{L}$.

Q: Method — simplest form of an age ratio: Azhagan is 50 years old, his son is 10. What is the simplest ratio of their ages?

A: $50:10$ reduces (divide both by 10) to $5:1$.

Q: Method — finding an unknown in a proportion (cross-multiplication): If the numbers 2, 5, x , 20 (in that order) are in proportion, find x .

A: In proportion, $2:5 = x:20$, so $2 \times 20 = 5 \times x \rightarrow 40 = 5x \rightarrow x = 8$. (General rule: in $a:b=c:d$, $a \times d = b \times c$.)

Q: Method — the '4th proportional': Find the 4th proportional to 4, 16, and 7.

A: The 4th proportional x satisfies $4:16 = 7:x \rightarrow 4x = 16 \times 7 = 112 \rightarrow x = 28$.

Q: Method — splitting a total using a 3-part ratio and a given average: Three numbers are in ratio $3:5:7$ and their average is 60. Find the smallest number.

A: Average of 60 over 3 numbers means their total = 180. Split 180 in ratio $3:5:7$ (15 parts total): each part = $180/15 = 12$. Smallest = $3 \times 12 = 36$.

Q: Method — 'duplicate ratio' (squaring a ratio): What is the duplicate ratio of $2:7$?

A: The duplicate ratio of $a:b$ is $a^2:b^2$. So the duplicate ratio of $2:7$ is $4:49$.

MOCK TEST (WITH ANSWERS)

1. A woman spends Rs.620 buying 14 pens and notebooks. One pen costs Rs.30, one notebook costs Rs.50. Find the ratio of notebooks to pens purchased.

- A. 2:5
- B. 5:2 ✓**
- C. 3:4
- D. 4:3

Explanation: $p+n=14$; $30p+50n=620 \rightarrow 3p+5n=62$. Substitute $p=14-n$: $3(14-n)+5n=62 \rightarrow 42+2n=62 \rightarrow n=10$, $p=4$. Ratio notebooks:pens = $10:4 = 5:2$. [hard, authored]

2. If $x/8 = y/12 = z/20$, find $x:y:z$ in lowest terms.

- A. 2:3:5 ✓**
- B. 8:12:20
- C. 4:6:10
- D. 1:2:3

Explanation: $x:y:z = 8:12:20$, which reduces (divide by HCF=4) to $2:3:5$. [easy, authored]

3. If $P:Q = 2:3$ and $Q:R = 9:10$, find $P:R$.

A. 3:5 ✓

B. 2:5

C. 3:10

D. 9:20

Explanation: Scale $P:Q=2:3$ by 3 to get $P:Q=6:9$, matching $Q=9$ from $Q:R=9:10$. So $P:Q:R=6:9:10$, giving $P:R=6:10=3:5$. [medium, authored]

4. A 40 L mixture has milk:water = 3:1. How much water must be added to change the ratio to 3:2?

A. 5 L

B. 10 L ✓

C. 8 L

D. 6 L

Explanation: Milk:water=3:1 in 40L means milk=30L, water=10L. Milk stays fixed; we want water= $(2/3) \times \text{milk} = (2/3) \times 30 = 20$ L. Water to add = $20 - 10 = 10$ L. [hard, authored]

5. If 5, 8, x, 24 are in proportion, find x.

A. 12.5

B. 15 ✓

C. 16

D. 20

Explanation: $5:8 = x:24 \rightarrow 5 \times 24 = 8x \rightarrow 120 = 8x \rightarrow x = 15$. [easy, authored]

6. Find the 4th proportional to 3, 12, and 9.

A. 27

B. 36 ✓

C. 30

D. 33

Explanation: $3:12 = 9:x \rightarrow 3x = 12 \times 9 = 108 \rightarrow x = 36$. [easy, authored]

7. Three numbers are in the ratio 2:3:4 and their average is 45. Find the largest number.

A. 40

B. 50

C. 60 ✓

D. 45

Explanation: Average 45 over 3 numbers means total=135. 9 parts total, each part=15. Largest = $4 \times 15 = 60$. [medium, authored]

8. What is the duplicate ratio of 3:5?

A. 6:10

B. 9:25 ✓

C. 9:10

D. 3:25

Explanation: Duplicate ratio = square each term: $3^2:5^2 = 9:25$. [easy, authored]

9. A sum of Rs.450 is divided among 100 boys and girls, each boy getting Rs.5.10 and each girl Rs.3.60. Find the number of girls.

A. 20

B. 30

C. 40 ✓

D. 50

Explanation: Let girls=g, boys=100-g. $3.6g + 5.1(100-g) = 450 \rightarrow 3.6g + 510 - 5.1g = 450 \rightarrow -1.5g = -60 \rightarrow g = 40$. [hard, authored]

10. A father's age (40) to his daughter's age (8) — what is the simplest ratio?

A. 5:1 ✓

B. 8:1

C. 4:1

D. 10:1

Explanation: 40:8 reduces ($\div 8$) to 5:1. [easy, authored]

Part C – General English: Poems: "The Crocodile" (Lewis Carroll) & "From a Railway Carriage" (R.L. Stevenson)

READING MATERIAL

Two short poems bundled into one day. Both are tested almost entirely through figure-of-speech identification on specific lines (matching the syllabus's explicit framing for Unit VII), so the Q&A below pairs each notable line with its device, plus the background/context that 'appreciation' questions tend to probe.

Q: Where does "The Crocodile" actually come from, and is it an original poem?

A: It's a parody. Lewis Carroll wrote it as a parody of Isaac Watts' moralistic poem 'Against Idleness and Mischief' ('How doth the little busy bee...'). Carroll's version appears as a poem recited by Alice in Chapter 2 of Alice's Adventures in Wonderland, originally titled 'How Doth the Little Crocodile'.

Q: What is the crocodile doing with its tail, and what mood does it project?

A: It is improving/grooming its shining tail to make it shine more. The crocodile is portrayed as cunning and outwardly cheerful.

Q: Identify the figure of speech: "How cheerful he seems to grin."

A: Personification (and the line is also cited as an example of alliteration in some sources — 'cheerful' / 'seems' sound patterns) — attributing a human-like cheerful expression to the crocodile.

Q: Identify the figure of speech: "Golden scale" describing the crocodile's skin.

A: Metaphor — the scale isn't literally gold, it's being described AS gold for its gleam.

Q: What do the 'golden scales' suggest about the crocodile, beyond just appearance?

A: An attractive trap — the glittering, attractive appearance lures fish in, even though the crocodile's true intent is predatory. This is the poem's central irony.

Q: Identify the figure of speech: "And welcomes little fishes in... with gently smiling jaws."

A: Irony — the crocodile 'welcomes' the fish in a friendly-sounding way, but the true purpose is to eat them. The gap between the gentle/welcoming language and the predatory reality is the poem's comic-dark irony. ('Smiling jaws' on its own is also cited as personification — a gentle smiling expression is a human trait, here applied to an animal's mouth.)

Q: Why does the crocodile grin, according to the poem's logic?

A: To deceive the fish — the cheerful, welcoming appearance is a lure, not genuine friendliness.

Q: Why does the poet mention the River Nile?

A: To establish the crocodile's natural habitat — grounding the otherwise whimsical, talking-animal poem in a real geographical setting.

Q: What is the overall lesson/moral the poet conveys through "The Crocodile"?

A: "Be careful of appearances" — cheerful, welcoming surfaces can hide dangerous intentions, a theme reinforced by the poem itself being a parody of a much more sincere moral poem (Watts' bee poem about industriousness).

Q: What is the setting and origin of "From a Railway Carriage"?

A: It's narrated from a child's point of view, looking out of a train window at the rapidly passing countryside. It was published in Robert Louis Stevenson's 1885 collection A Child's Garden of Verses. The poem's galloping rhythm is written to mimic the rhythmic chugging/motion of a moving train.

Q: Identify the figure of speech: "Faster than fairies, faster than witches."

A: Simile (uses 'than' to compare the train's speed to fairies and witches) — and the same line is also cited as alliteration, for the repeated 'f'/'w' sounds across 'faster'/'fairies' and the repeated structure.

Q: Identify the figure of speech: "Bridges and houses, hedges and ditches."

A: Alliteration — paired repeated consonant sounds ('h' in houses/hedges, and the rhythmic pairing itself) that mimic the train's clattering rhythm.

Q: Identify the figure of speech: "Charging along like troops in a battle" and "Fly as thick as driving rain."

A: Both are similes — direct comparisons using 'like'/'as', describing how things outside the train rush past.

Q: Identify the figure of speech: "And ever again, in the wink of an eye."

A: Metaphor (an idiom used metaphorically) — 'wink of an eye' isn't literal, it stands for an extremely brief moment.

Q: Identify the figure of speech: "Painted stations whistle by."

A: This line does double duty: Personification (stations don't literally 'whistle' — a human/animal-like action is given to an inanimate object) AND Onomatopoeia ('whistle' is itself a sound-word). Different MCQs may ask for either device on this exact line.

Q: Identify the figure of speech: "Here is a child who clammers and scrambles" and "Lumping along with man and load."

A: Both are cited as consonance — repetition of matching consonant sounds at the end of nearby words (clammers/scrambles; lumping/man/load), distinct from alliteration which repeats STARTING sounds.

MOCK TEST (WITH ANSWERS)

1. "The Crocodile" by Lewis Carroll is a parody of which earlier poem?

A. Wordsworth's 'Daffodils'

B. Isaac Watts' 'Against Idleness and Mischief' ✓

C. Blake's 'The Tyger'

D. Stevenson's 'Happy Thought'

Explanation: It parodies Isaac Watts' moralistic 'Against Idleness and Mischief' ('How doth the little busy bee'). [hard, authored]

2. In which work does Lewis Carroll's crocodile poem originally appear?

A. Through the Looking-Glass

B. Alice's Adventures in Wonderland ✓

C. The Hunting of the Snark

D. Sylvie and Bruno

Explanation: Alice recites it in Chapter 2 of Alice's Adventures in Wonderland. [hard, authored]

3. "Golden scale" describing the crocodile's skin is an example of:

A. Simile

B. Metaphor ✓

C. Onomatopoeia

D. Hyperbole

Explanation: Metaphor — directly calling the scale 'golden' rather than comparing with 'like/as'. [medium, web-research]

4. The crocodile 'welcoming' little fishes in, only to eat them, is best described as an example of:

A. Simile

B. Hyperbole

C. Irony ✓

D. Euphemism

Explanation: Irony — the gap between the friendly 'welcome' and the predatory reality. [medium, web-research]

5. What overall lesson does "The Crocodile" convey?

A. Always swim in groups

B. Be careful of appearances ✓

C. Fish are smarter than crocodiles

D. Rivers are dangerous

Explanation: A cheerful, welcoming surface can hide dangerous intent. [easy, web-research]

6. "From a Railway Carriage" was published in which Stevenson collection?

A. Treasure Island

B. A Child's Garden of Verses ✓

C. Kidnapped

D. The Black Arrow

Explanation: Published in A Child's Garden of Verses (1885). [hard, authored]

7. "Faster than fairies, faster than witches" is an example of:

A. Personification

B. Simile ✓

C. Onomatopoeia

D. Oxymoron

Explanation: A simile, comparing the train's speed using 'than'. [easy, web-research]

8. "Charging along like troops in a battle" uses which figure of speech?

A. Metaphor

B. Simile ✓

C. Personification

D. Hyperbole

Explanation: Simile — direct comparison using 'like'. [easy, web-research]

9. "Painted stations whistle by" — which figure of speech gives stations a human/animal-like action?

A. Onomatopoeia

B. Personification ✓

C. Simile

D. Alliteration

Explanation: Personification — stations can't literally whistle. [medium, web-research]

10. In the same line, "whistle by" is also an example of which device, since 'whistle' imitates a sound?

A. Onomatopoeia ✓

B. Metaphor

C. Consonance

D. Hyperbole

Explanation: Onomatopoeia — 'whistle' is a sound-imitating word. [medium, web-research]

11. "Here is a child who clambers and scrambles" exhibits which sound device?

A. Alliteration

B. Consonance ✓

C. Assonance

D. Onomatopoeia

Explanation: Consonance — matching consonant sounds at the end of nearby words. [medium, web-research]

12. "And ever again, in the wink of an eye" uses 'wink of an eye' as a:

A. Simile

B. Metaphor ✓

C. Personification

D. Onomatopoeia

Explanation: Metaphor — a brief-moment idiom used non-literally. [medium, web-research]